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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(SL2, BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks: 25

Comparative Analysis

83%

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

	Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
B	Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
C	Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
D	Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
E	Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
F	Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
G	Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
H	Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

41.5

	A	B	C	D	E	F	G	H	Total
Q ₁	1.5	2	1	0.5	1	1	1		9
Q ₂	1	2	1	1	1	1	1	0.5	8.5
Q ₃	1.5	1.5	0.5	1.5	1	1	1	0.5	8.5
Q ₄	1	2	0.5	1	1	0.5	1	0.5	7.5
Q ₅	1.5	2	0.5	1.5	0.5	0.5	1	0.5	8

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Comparative Analysis

Code of Conduct:

I __j.manoj(192572143)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

j.manoj

Question 1:

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Introduction & Course Objectives:

Network topology is the arrangement of various elements (links, nodes, etc.) of a computer network. Choosing the correct topology is critical for optimizing performance, managing costs, ensuring fault tolerance, and streamlining network administration.

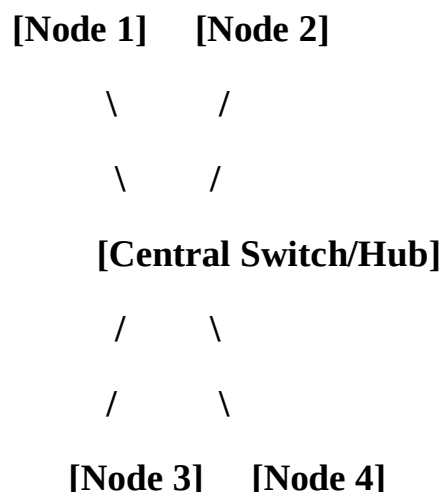
Objective:

To compare Star Topology and Bus Topology in terms of Cost, Reliability, and Performance.

PDF

Topology Diagrams

1.Star Topology Diagram:



2.Bus Topology Diagram:

Plaintext

[Terminator]=== [Node 1] === [Node 2] === [Node 3] === [Terminator]

Comparative Analysis Table:

Parameter	Star Topology	Bus Topology
Primary Device	Requires a central hub or switch	Uses a single shared backbone cable
Cost	Generally more expensive due to extra cabling and central hardware	Low-cost due to minimal cabling requirements
Reliability	Superior; failure of one node does not impact others	Lower; failure of the main bus cable brings down the entire network
Performance	High; managed via central device, reducing data collisions	Degrades as devices increase due to heavy traffic on shared medium

Detailed Explanation:

Cost Analysis

Star topology incurs higher expenses because every workstation must be connected independently to a central networking device (such as an Ethernet switch or hub), requiring significant cabling. Conversely, bus topology utilizes a single linear backbone cable that runs through the network, making the initial setup economical and straightforward.

Reliability Analysis

In a star configuration, if an individual cable or node fails, the rest of the network operates seamlessly, rendering it highly reliable. In contrast, a bus network presents a single point of failure: if the main backbone cable breaks or a terminator is damaged, the communication channel is severed globally, crashing the entire network.

Performance Analysis

Data transmission in a star network is regulated through the central switch, which optimizes traffic flow and minimizes packet collisions. A bus network relies on a shared communication medium; as more devices are added, data traffic escalates, leading to frequent collisions and noticeable performance degradation.

Advantages and Disadvantages:

- Star Topology Advantages:** Easy to add/remove devices; excellent fault isolation; high performance.

- **Star Topology Disadvantages:** Higher hardware cost; central device dependency (if the switch fails, the whole network goes down).
- **Bus Topology Advantages:** Low implementation cost; easy to set up in small environments; requires minimal cable length.
- **Bus Topology Disadvantages:** Difficult troubleshooting; whole network failure upon cable rupture; severe performance bottlenecks under heavy loads.

Real-World Applications:

- **Star Topology:** Modern office Local Area Networks (LANs), home Wi-Fi networks connected to a wireless router.
- **Bus Topology:** Legacy Ethernet networks (10Base2 / ThinNet) and small-scale experimental setups where cost is the primary constraint.

Conclusion:

While bus topology provides an affordable and simple solution for minimal setups, star topology is vastly superior in performance, scalability, and fault isolation, making it the industry standard for modern enterprise networking.

Question 2:

Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Introduction & Course Objectives:

Network topology is the arrangement of various elements (links, nodes, etc.) of a computer network. Choosing the correct topology is critical for optimizing performance, managing costs, ensuring fault tolerance, and streamlining network administration.

Objective:

To compare Mesh Topology and Star Topology based on Fault Tolerance and Reliability.

Topology Diagrams:

1.Mesh Topology Diagram:

[Node 1] ----- [Node 2]

| \ / |
| \ / |
| \ / |
| \ / |

[Node 3] ----- [Node 4]

2.Star Topology Diagram:

[Node 1] [Node 2]

 \ /
 \ /

[Central Switch/Hub]

 / \
 / \

[Node 3] [Node 4]

Comparative Analysis Table:

Parameter	Mesh Topology	Star Topology
Connectivity	Every node is connected to multiple other nodes PDF	Nodes connect radially to a single central hub/switch PDF
Fault Tolerance	Very high; multiple redundant paths available for routing PDF	Moderate; individual node failures are isolated, but central hub failure is fatal PDF
Reliability	Exceptionally robust for critical environments PDF	High for node links, but vulnerable to core device failure

Parameter	Mesh Topology	Star Topology
Network Availability	Near 100% uptime due to path redundancy	High uptime, contingent on central hardware stability

Detailed Explanation:

Fault Tolerance Analysis

Mesh topology delivers near-absolute fault tolerance because every node maintains direct point-to-point links with several other nodes. If a primary link is severed, routing protocols instantly redirect traffic through alternate paths without disrupting active communication. Star topology offers moderate fault tolerance: while individual client disconnections do not affect other workstations, any failure of the central hub or switch halts the entire network.

Reliability & Network Availability Discussion

In mission-critical environments where downtime is unacceptable (such as financial trading floors, air traffic control, or military command centers), mesh topology is preferred due to its redundancy. Star topology provides dependable reliability for everyday business operations, but it inherently relies on the operational health of its central switching hardware.

Advantages and Disadvantages:

- **Mesh Topology Advantages:** Maximum redundancy; secure direct data transfer; failure isolation without network collapse.
- **Mesh Topology Disadvantages:** Immense cabling complexity; high deployment and port costs.
- **Star Topology Advantages:** Cost-effective compared to mesh; straightforward troubleshooting; easier expansion up to port limits.
- **Star Topology Disadvantages:** Vulnerability to central device failure.

Industry Examples

- **Mesh Topology:** Backbone WANs, military defense networks, and critical IoT sensor grids.
- **Star Topology:** Corporate enterprise floor networks, institutional computer labs, and VoIP telephone systems.

Conclusion:

Mesh topology outclasses star topology in environments demanding zero tolerance for failure, whereas star topology strikes an optimal balance between reliability and cost-efficiency for standard enterprise requirements.

Question 3:

Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Introduction & Course Objectives:

Network topology is the arrangement of various elements (links, nodes, etc.) of a computer network. Choosing the correct topology is critical for optimizing performance, managing costs, ensuring fault tolerance, and streamlining network administration.

Objective

To compare Bus Topology and Mesh Topology based on Installation Complexity.

Topology Diagrams

Bus Topology Diagram:

Plaintext

[Terminator]==== [Node 1] === [Node 2] === [Node 3] === [Terminator]

Mesh Topology Diagram:

Plaintext

[Node 1] ----- [Node 2]

| \ / |

| \ / |

| \ / |

| \ / |

[Node 3] ----- [Node 4]

Installation Requirements Table

Feature	Bus Topology	Mesh Topology
Cabling Requirement	Single continuous backbone cable with minimal drop lines PDF	Massive amount of dedicated cables connecting every node pair PDF
Port Requirements	Minimal interface ports per node	High number of NICs/ports required per node ($n - 1$ ports)
Installation Time	Fast and simple; ideal for temporary setups PDF	Time-consuming, intricate, and labor-intensive PDF
Financial Outlay	Economical PDF	Extremely expensive PDF

Detailed Explanation

Cabling and Complexity Analysis

Bus topology is celebrated for its structural simplicity. It uses a single backbone cable where devices tap directly into the line, minimizing physical clutter and installation overhead. Conversely, mesh topology requires a complex web of dedicated point-to-point links. For n nodes, a full mesh requires $\frac{n(n-1)}{2}$ physical links, causing installation complexity and cable management requirements to scale exponentially.

Cost and Deployment Comparison

Because of its simple cabling layout, bus topology requires minimal technical expertise and lower material costs to deploy. Mesh topology demands specialized network engineering, extensive physical cabling, and hardware equipped with multiple network interface cards (NICs), making it significantly costlier and harder to implement.

Advantages and Disadvantages

- **Bus Topology Advantages:** Minimal wire usage; quick installation; low financial barrier.
- **Bus Topology Disadvantages:** Difficult to isolate breaks; poor scalability.
- **Mesh Topology Advantages:** Complete redundancy; robust fault isolation; maximum security.
- **Mesh Topology Disadvantages:** Extreme installation complexity; prohibitive wiring costs.

Real-World Examples

- **Bus Topology:** Small legacy classroom setups or temporary testing networks.
- **Mesh Topology:** Internet Service Provider (ISP) core backbones and high-security government communication grids.

Conclusion

When ease of installation and budget are paramount, bus topology is vastly superior. However, when structural resilience outweighs installation effort, mesh topology remains the definitive choice despite its complexity.

Question 4:

Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Introduction & Course Objectives:

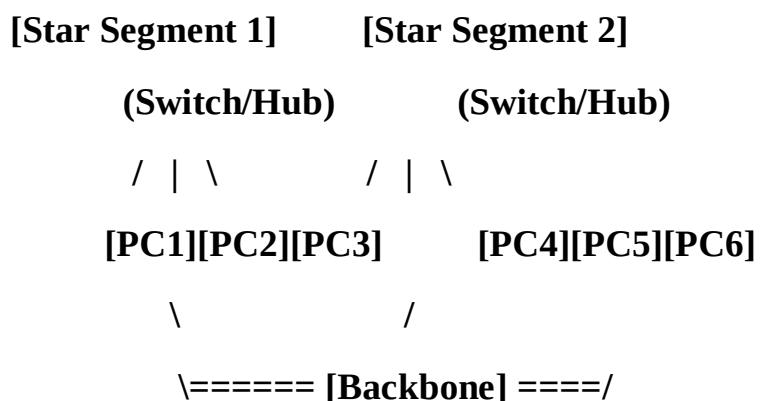
Network topology is the arrangement of various elements (links, nodes, etc.) of a computer network. Choosing the correct topology is critical for optimizing performance, managing costs, ensuring fault tolerance, and streamlining network administration.

Objective:

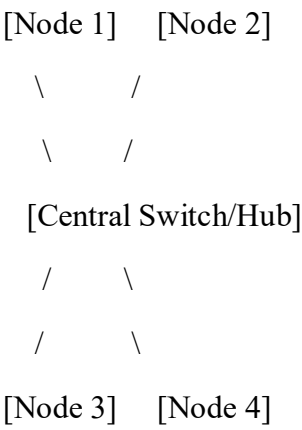
To compare Hybrid Topology and Star Topology based on Scalability.

Topology Diagrams

1. Hybrid Topology Diagram (Star-Ring Combination):



2.Star Topology Diagram:



Scalability Comparison Table:

Parameter	Hybrid Topology	Star Topology
Architecture	Integration of two or more distinct topologies (e.g., star-ring, star-bus) PDF	Centralized connection radiating outward from a single hub/switch PDF
Expansion Capability	Highly flexible; new segments can be added effortlessly without restructuring PDF	Limited by the port capacity of the central device PDF
Infrastructure Adaptation	Easily customized to fit building layouts and departmental needs	Rigidly bound to central switch locations

Detailed Explanation:

Scalability and Expansion Analysis

Hybrid topology provides unmatched scalability because it merges multiple topologies, allowing network administrators to expand infrastructure organically to match organizational growth. New functional departments can integrate distinct sub-networks (like a star or bus segment) into the existing architecture without interrupting active services.

In contrast, star topology has a hard ceiling on scalability dictated by the port count of its central switch or hub. Once all ports are occupied, expanding the network requires procuring additional hardware, cascading switches, or restructuring cable runs.

Infrastructure Requirements

Hybrid networks accommodate heterogeneous hardware and legacy systems by tying disparate topologies together seamlessly. Star topologies demand uniform cabling (typically twisted-pair Ethernet) converging on central wiring closets.

Advantages and Disadvantages:

- **Hybrid Topology Advantages:** Extremely scalable; highly customizable; fault isolation localized to sub-segments.
- **Hybrid Topology Disadvantages:** Complex design and management; expensive hardware integration.
- **Star Topology Advantages:** Easy to manage; predictable expansion within port limits.
- **Star Topology Disadvantages:** Port bottlenecks; single point of failure at the core switch.

Enterprise Use Cases

- **Hybrid Topology:** Large multi-building corporate campuses, university networks uniting department-specific LANs.

Star Topology: Small-to-Medium Enterprise (SME) floor offices and retail store local networks.

Conclusion:

Hybrid topology offers superior long-term scalability for growing enterprises by blending diverse structural designs, whereas star topology is best suited for stable, bounded operational environments.

Question 5:

In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Introduction & Course Objectives:

Network topology is the arrangement of various elements (links, nodes, etc.) of a computer network. Choosing the correct topology is critical for optimizing performance, managing costs, ensuring fault tolerance, and streamlining network administration.

Objective:

To compare the Maintenance Requirements of Star, Mesh, Bus, and Hybrid Topologies.

Comparative Table:

Topology	Maintenance Complexity	Troubleshooting Ease	Administration Skill Level Required
Star Topology	Low to Moderate	High; issues are isolated quickly to specific nodes/cables PDF	Basic to Moderate
Mesh Topology	High	Difficult due to intricate cross-connections and massive cabling PDF	Advanced
Bus Topology	Moderate to High	Hard; backbone faults are difficult to pinpoint and disrupt all nodes PDF	Moderate
Hybrid Topology	Moderate to High	Dependent on sub-segment design; requires rigorous monitoring PDF	Advanced

Detailed Explanation & Troubleshooting Methods:

Maintenance Analysis

Maintaining a computer network involves routine health checks, cable management, hardware upgrades, and fault resolution.

- **Star Topology:** Relatively easy to maintain. If a cable or client machine malfunctions, management software or switch port indicators isolate the fault instantly without affecting the rest of the network.
- **Mesh Topology:** Highly reliable in operation, but notoriously difficult to maintain. Tracing faults across a dense web of redundant links requires advanced cable tracing tools and meticulous documentation.

- **Bus Topology:** Troublesome to maintain. A break anywhere along the linear backbone brings down the entire network, and locating the exact point of failure on a long cable run requires specialized Time Domain Reflectometers (TDRs).
- **Hybrid Topology:** Requires balanced administrative oversight. While individual segments can be maintained independently, managing the overall inter-topology bridges demands skilled technicians and structured network monitoring tools.

Advantages and Disadvantages

- **Star Maintenance:** Quick isolation; simple management tools.
- **Mesh Maintenance:** Complex diagnostics; labor-intensive upkeep.
- **Bus Maintenance:** Prone to hidden breakages; frustrating diagnostic processes.
- **Hybrid Maintenance:** Flexible isolation, but requires expert administrative oversight.

Industry Examples & Summary:

- **Star:** Corporate offices utilizing SNMP-managed switches for remote diagnostics.
- **Mesh:** Telecom carrier core networks requiring dedicated Network Operations Center (NOC) teams.
- **Bus:** Legacy manufacturing floor terminals.
- **Hybrid:** Global enterprise networks uniting regional star LANs with mesh WAN backbones.

Conclusion:

Implementing VLAN architecture within a structured multi-department enterprise network optimizes broadcast traffic, hardens security boundaries, and provides seamless scalability for future corporate expansion